Week 13 — Lecture notes

Course review

The web version of these notes is at tchaffey.com/elec2302.

Outline:

1. Course overview (a non-exhaustive recap).
2. Where next? Survey of applications.
   - feedback control (X-29 available bandwidth) (quadcopters & grid frequency regulation)
   - Audio (noise cancellation & echo cancellation)
   - Graphics (spatial convolution, 2D convolution, 2D Fourier transform, CNNs and paper mill control, MRI).
   - Filter banks and the Cochlear implant.
   - Beyond LTI & sinusoids: wavelets, describing functions, SRGs
3. Follow on courses: DSP, control, power systems, communications, electronics.
4. The final exam.
   - structure
   - practice exam questions.

Course review

Four parts:

1. Time domain signals and systems
2. Frequency domain
3. Stability & Laplace
4. Applications: filtering, modulation, system ID.

1. Time domain

A system that is linear satisfies superposition: if $u_1 \mapsto y_1$, $u_2 \mapsto y_2$, then

$$\alpha u_1 + \beta u_2 \;\mapsto\; \alpha y_1 + \beta y_2.$$

If the system is time invariant, then $u(t-\tau) \mapsto y(t-\tau)$.

We also have a special signal δ such that every input is a sum of δ's:

$$u(t) = \int_{-\infty}^{\infty} u(\tau)\,\delta(\tau - t)\,\mathrm{d}\tau. \qquad \text{(The sifting property)}.$$

So if we know the response of the system to a δ (i.e. the impulse response), the response to any other input is a weighted sum of shifted impulse responses:

$$\boxed{y(t) = \int_{-\infty}^{\infty} u(t-\tau)\,h(\tau)\,\mathrm{d}\tau.}$$

This is the convolution.

It explains a big mystery of the course: we can use exactly the same tools to study signals and systems because an LTI system is completely described by a single signal: the impulse response.

Pro: only need a single response to characterise the system.

Con: convolution is difficult to calculate.

2. Frequency domain

The reason that convolution is difficult is that the response at time t is a combination of responses to inputs at every time before. But it turns out we can do a change of basis and index by frequency instead of time:

- Any signal is a sum of sinusoids
- The output at frequency ω depends only on the input at frequency ω.

This turns convolution into multiplication.

Periodic signals: if the period is T, the signal is a sum of sinusoids at frequencies $k\dfrac{2\pi}{T}$, $k \in \mathbb{Z}$. This is the Fourier series:

$$\boxed{u(t) = \sum_k c_k \, e^{jk\omega_0 t}, \qquad \omega_0 = \frac{2\pi}{T}.}$$

If a signal is aperiodic, the sum becomes an integral and we have a transform:

$$\boxed{u(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} U(j\omega) \, e^{j\omega t} \, \mathrm{d}\omega.}$$

c_k & $U(j\omega)$ are computed by projecting the signal onto a sinusoid of frequency $k\omega_0$ / ω:

$$\boxed{U(j\omega) = \int_{-\infty}^{\infty} u(t) \, e^{-j\omega t} \, \mathrm{d}t.}$$

The same method can also be used to filter noise out of a sinusoidal signal when doing system ID.

The power, above all, of the Fourier transform is that convolution becomes multiplication:

$$u(t) * y(t) \;\longleftrightarrow\; U(j\omega)\, Y(j\omega).$$

This lets us talk about systems as filters that multiply and phase shift their input signals. We can design filters to remove unwanted parts of the spectrum, like noise and modulation artefacts. A common filter type is Butterworth, which has frequency response gain

$$\frac{1}{1 + \left(\dfrac{\omega}{\omega_c}\right)^{2n}} = |G(j\omega)|^2.$$

Modulation lets us shift the spectrum of a signal:

$$u(t)\cos(\omega_0 t) \;\xrightarrow{\text{F.T.}}\; \tfrac{1}{2}U(j(\omega - \omega_0)) + \tfrac{1}{2}U(j(\omega + \omega_0)).$$

To demodulate such a signal, multiply again by the carrier:

$$\begin{aligned} u(t)\cos^2(\omega_0 t) &= \tfrac{1}{2}u(t) + \tfrac{1}{2}u(t)\cos(2\omega_0 t) \\ &\rightsquigarrow\; \tfrac{1}{2}U(j\omega) + \underbrace{\tfrac{1}{4}U(j(\omega - 2\omega_0)) + \tfrac{1}{4}U(j(\omega + 2\omega_0))}_{\text{These terms can be filtered out.}} \end{aligned}$$

3. Stability & Laplace

The Fourier transform exists only when the integral

$$\int_{-\infty}^{\infty} u(t)\, e^{-j\omega t}\, \mathrm{d}t$$

converges.

If it doesn't converge, for example if $u(t) = e^{\alpha t}$ (the impulse response of an unstable system) we can shift the axis of integration:

$$\int_{-\infty}^{\infty} u(t)\, e^{(-j\omega-\sigma)t}\, \mathrm{d}t = U(s) \quad \text{where } s = j\omega + \sigma.$$

The decaying term $e^{-\sigma t}$ cancels growing terms in $u(t)$. This is the bilateral Laplace transform.

The unilateral Laplace transform integrates from $t = 0$ and takes into account initial conditions.

If we have a differential equation

$$\ddot{y} + \alpha\dot{y} + \beta y = \gamma u$$

we can take the Laplace transform of each side to get

$$s^2Y(s) - sy(0) - \dot{y}(0) + \alpha sY(s) - \alpha y(0) + \beta Y(s) = \gamma U(s).$$

If we set all initial conditions to zero, and $u(t) = \delta(t)$, $U(s) = 1$ and we get the Laplace transform of the impulse response, which is the transfer function:

$$s^2Y(s) + \alpha sY(s) + \beta Y(s) = \gamma$$

$$Y(s) = \frac{\gamma}{s^2 + \alpha s + \beta}.$$

The zeros of the denominator are the poles of the system. If they are in the LHP, the system is exponentially stable. A pole in the RHP gives instability.

The coefficients tell us about the behaviour of the system:

$$\frac{k\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ω_n: natural frequency (frequency of oscillation)

ζ: damping ratio (how much damping)

k: DC gain.

Measuring these values allows us to estimate a transfer function from data.

Separating a Laplace transform into partial fractions lets us calculate the inverse Laplace transform by matching terms in the table.